`nmr_extract` Operator Use Guide

Generated/Reviewed Documentation (Codex/Opus/Human) for v0.0.1

1 What the extractor does

`nmr_extract` builds a charged protein model from one of five input shapes, runs the calculator stack for that model, and writes NumPy feature arrays. In trajectory mode it also streams GROMACS frames, writes per-frame outputs, and writes a trajectory HDF5 file.

APBS and AIMNet2 are part of the run in every mode. They are not controlled by `-no-apbs`, `-no-coulomb`, or any other switch. MOPAC is the available expensive-method toggle, but the current parser implements it asymmetrically: modes 1–4 default MOPAC on and accept `-no-mopac`; mode 5 defaults MOPAC off and accepts `-mopac`.

2 Command shape

Every run needs exactly one mode selector:

```
--pdb FILE
--protonated-pdb FILE
--orca --root NAME
--mutant --wt NAME --ala NAME
--trajectory DIR
```

The parser rejects a command with no mode:

```
no mode flag found
  valid modes: --pdb, --protonated-pdb, --orca, --mutant, --trajectory
```

It also rejects more than one mode selector:

```
multiple mode flags present: MODE1 and MODE2 (pick exactly one)
```

`-help`, `-h`, or no arguments prints usage and exits without running. File existence is checked after parsing. For a required file, the path must exist and must be a regular file. For a required directory, the path must exist and must be a directory.

Unknown flags are not a supported interface. The current parser does not have a general unknown-flag rejection pass, so an unknown flag can be ignored when a valid mode is otherwise present. Do not rely on that behavior.

3 Common options

`-output DIR` is required for every successful run. The directory may already exist. If it exists, it must be a directory:

```
--output: exists but is not a directory: DIR
```

If -output is omitted, execution stops with:

```
ERROR: --output DIR required
```

-config FILE is optional. It points to a TOML file with calculator parameter overrides. If provided, validation requires it to be a regular file:

```
--config: file not found: FILE
--config: not a regular file: FILE
```

If -config is not provided, the executable tries NMR_DATA_DIR/calculator_params.toml when that file exists.

-aimnet2 FILE is an AIMNet2 model path, not an AIMNet2 on/off switch. AIMNet2-derived calculators require a model. The resolution order is: the CLI path, then NMR_AIMNET2_MODEL, then aimnet2_model_path from the loaded calculator TOML. A TOML-relative model path is resolved relative to that TOML file. If a CLI or resolved path is present, validation requires a regular file:

```
--aimnet2 model: file not found: FILE
--aimnet2 model: not a regular file: FILE
```

If no model path can be resolved, execution stops before any mode runner:

```
ERROR: AIMNet2 model required: pass --aimnet2 PATH, set NMR_AIMNET2_MODEL, or set
    aimnet2_model_path in calculator_params.toml.
```

4 Mode 1: bare PDB

Use this mode for a PDB that should be protonated by reduce at runtime.

```
nmr_extract --pdb input.pdb --output out --aimnet2 model.jpt
```

Optional flags:

- -pH VALUE: parsed as a floating-point value with default 7.0. Current execution records this value in the build context and log. The ProtonateWithReduce implementation has no pH parameter, so this value does not alter reduce's global settings in the current code.
- -no-mopac: disables MOPAC-family single-conformation calculators. Without this flag, MOPAC is on in this mode.

Required input:

- FILE after -pdb. Validation rejects an omitted path, a missing file, or a non-regular file.

Parse and validation diagnostics include:

```
--pdb requires a file path: --pdb FILE
--pdb: file not found: FILE
--pdb: not a regular file: FILE
```

The reader loads the original file, calls `reduce` in BUILD mode, parses the protonated PDB with cif++, assigns ff14SB charges, prepares force-field charges, and computes the net charge by rounding the charge sum to the nearest integer.

The reduce wrapper strips existing ATOM hydrogens before rebuilding them. It adds HIS sidechain hydrogens, keeps OH/SH hydrogens, adds non-water hydrogens, does not add water hydrogens, rotates existing OH groups, flips NQH groups, and runs reduce optimization. The het dictionary must exist at the CMake-provided `HET_DICTIONARY` path or the `REDUCE_HET_DICT` override path. If the input text is empty, no model is parsed, the het dictionary is missing, or reduce returns no PDB text, this mode fails.

PDB parsing rules:

- `TITLE`, `REMARK`, `MODEL`, `ENDMDL`, and `USER` records are skipped before cif++ parsing.
- A blank chain ID in an ATOM or HETATM record is changed to `A`.
- An `END` record is appended if the input did not contain one.
- Only alternate locations with an empty alt id or `A` are loaded.
- Unknown elements are skipped.
- Residues that do not map to a known amino-acid type are skipped.
- If no protein atoms remain, parsing fails with `No protein atoms found in PDB input`.
- Non-numeric residue sequence IDs are logged and stored as sequence number `0`; they are not a hard rejection in this reader.

Charge setup uses `RuntimeEnvironment::Ff14sbParams()`. If the ff14SB flat table cannot be found, the mode fails with an ff14SB params error. If charge resolution or charge preparation fails, the mode fails with that charge error.

Output:

- `atoms_category_info.npy`.
- Topology sidecars: `residues.npy`, `bonds.npy`, `rings.npy`, `ring_membership.npy`, and `extraction_manifest.json`.
- The identity arrays `pos.npy`, `element.npy`, `residue_index.npy`, and `residue_type.npy`.
- One set of feature arrays for the ConformationResults that attached in the run. APBS and AIMNet2 arrays are expected when their calculators succeed. MOPAC-family arrays are absent when `-no-mopac` is used.

5 Mode 2: already-protonated PDB

Use this mode for a PDB that already contains the intended hydrogen atoms.

```
nmr_extract --protonated-pdb protonated.pdb --output out --aimnet2 model.jpt
```

Optional flag:

- `--no-mopac`: disables MOPAC-family single-conformation calculators. Without this flag, MOPAC is on in this mode.

Required input:

- `FILE` after `--protonated-pdb`. Validation rejects an omitted path, a missing file, or a non-regular file.

Parse and validation diagnostics include:

```
--protonated-pdb requires a file path: --protonated-pdb FILE
--protonated-pdb: file not found: FILE
--protonated-pdb: not a regular file: FILE
```

This mode skips reduce. It parses the supplied PDB directly with the same PDB rules as mode 1, assigns ff14SB charges, prepares force-field charges, and writes the same output surface as mode 1.

6 Mode 3: ORCA/AMBER-prepared pose

Use this mode for a single tleap/AMBER-prepared pose described by a root name.

```
nmr_extract --orca --root runs/1abc/WT --output out --aimnet2 model.jpt
```

Optional flag:

- `--no-mopac`: disables MOPAC-family single-conformation calculators. Without this flag, MOPAC is on in this mode.

Required by CLI validation:

- `NAME.xyz`: protonated coordinates.
- `NAME.prmtop`: AMBER topology and charges.

The parser expands `--root NAME` exactly as:

```
NAME.xyz
NAME.prmtop
NAME_nmr.out
```

There is no globbing and no directory search. The `NAME_nmr.out` file is marked optional by CLI validation, but the current runner always supplies the expanded path to `OperationRunner`. A missing `NAME_nmr.out` therefore causes the run to fail when ORCA shielding is loaded:

```
NMR output not found: NAME_nmr.out
ORCA shielding load failed for NAME_nmr.out
```

For a first-time operator run, prepare `NAME_nmr.out` alongside the `.xyz` and `.prmtop`.

Parse and validation diagnostics include:

```
--orca requires --root NAME
  root expands to: {root}.xyz, {root}.prmtop, {root}_nmr.out
--orca .xyz: file not found: NAME.xyz
--orca .prmtop: file not found: NAME.prmtop
```

ORCA/AMBER reader rules:

- The XYZ first line must contain a positive atom count.
- The number of XYZ atoms must equal the number of atoms in the prmtop.
- The prmtop must contain `ATOM_NAME`, `RESIDUE_LABEL`, and `RESIDUE_POINTER`; otherwise the reader reports `incomplete prmtop`.
- Residue labels must resolve through the naming registry. Unknown residues are hard errors.
- The prmtop provides the atom names, residue labels, charges, and protonation variants. It provides elements when `ATOMIC_NUMBER` is present; otherwise the loader takes elements from the XYZ. The XYZ provides positions.
- The prmtop charge source later requires `CHARGE` and `RADII` rows for every protein atom.
- ORCA NMR output, when loaded, must contain the same number of nuclei as the conformation has atoms, and the ORCA nucleus element order must match the protein atom element order. A mismatch is a hard error because tensors would attach to the wrong atoms.

Output is the same single-conformation output surface as modes 1–2, with ORCA shielding arrays included when the ORCA shielding result attaches.

7 Mode 4: WT–ALA mutant pair

Use this mode for a WT pose and its alanine mutant pose. Each side is a mode-3 ORCA/AMBER-prepared pose.

```
nmr_extract --mutant --wt runs/1abc/WT --ala runs/1abc/ALA --output out --aimnet2 model.jpt
```

Optional flag:

- `-no-mopac`: disables MOPAC-family single-conformation calculators on both sides. Without this flag, MOPAC is on for both sides.

The parser expands each root independently:

```
WT root:
  WT_NAME.xyz
  WT_NAME.prmtop
  WT_NAME_nmr.out

ALA root:
  ALA_NAME.xyz
  ALA_NAME.prmtop
  ALA_NAME_nmr.out
```

CLI validation requires the `.xyz` and `.prmtop` file for each side and does not require either `_nmr.out` file. Current execution does pass both expanded `_nmr.out` paths to the ORCA shielding loader. If either file is missing or does not match its protonated geometry, the WT or ALA side fails and no mutant comparison output is produced.

Parse and validation diagnostics include:

```
--mutant requires --wt NAME --ala NAME
  each root expands to: {root}.xyz, {root}.prmtop, {root}_nmr.out
--mutant --wt .xyz: file not found: WT_NAME.xyz
--mutant --wt .prmtop: file not found: WT_NAME.prmtop
--mutant --ala .xyz: file not found: ALA_NAME.xyz
--mutant --ala .prmtop: file not found: ALA_NAME.prmtop
```

The runner loads both poses, runs the standard calculator sequence on both, then computes WT–ALA mutation delta tensors when both conformations have an ORCA shielding result. The output directory receives WT topology sidecars and WT conformation feature arrays. The mutation delta result attaches to the WT conformation; the runner does not write a separate full ALA output tree.

8 Mode 5: GROMACS trajectory

Use this mode for a GROMACS production directory with fixed file names.

```
nmr_extract --trajectory /data/1abc/production_scan1 --stride 10 --output out --aimnet2 model.jpt
```

MOPAC is off by default in this mode. Add `-mopac` to use the FullFat trajectory shape:

```
nmr_extract --trajectory /data/1abc/production_scan1 --stride 10 --mopac --output out --aimnet2
    model.jpt
```

The mode directory must contain these exact regular files:

```
DIR/production.tpr
DIR/production.trr
DIR/production.edr
```

The reader also looks for:

```
DIR/../topol.top
```

`topol.top` is not required by CLI validation. If it is present and its GROMACS-to-AMBER readback block parses, it is used to preserve residue chemistry decisions such as CYX and HIS variants. If it is missing or cannot be parsed, the loader falls back to the naming registry and logs that CYX/HIS resolution may be lost.

There is no file discovery. The derivation function is `FromProductionDir`, and it returns these paths:

```
return {dir / "production.tpr",
        dir / "production.trr",
        dir / "production.edr"};
```

Validation checks those exact paths:

```
e = CheckFile(files.tpr, "trajectory production.tpr"); if (!e.empty()) return e;
e = CheckFile(files.trr, "trajectory production.trr"); if (!e.empty()) return e;
e = CheckFile(files.edr, "trajectory production.edr"); if (!e.empty()) return e;
```

An `.xtc` file is not used. There is no separate explicit `production.xtc` rejection branch; a directory containing only `production.xtc` for coordinates fails because `production.trr` is absent:

```
trajectory production.trr: file not found: DIR/production.trr
```

The frame reader opens and reads `production.trr` through the TRR API:

```
gmx_trr_open
gmx_trr_read_frame
```

It reads positions, velocities when present, and the box matrix. A TRR without velocities is not rejected by this reader; the velocity buffer is left empty for velocity-consuming results. A TRR frame read for processing whose atom count differs from the TPR topology is rejected:

```
TRR frame natoms mismatch: header=N expected=M
```

8.1 Trajectory stride

The current code has one trajectory cadence knob:

```
--stride N
```

Default `N` is `1`. The parser converts `0` to `1`. Non-numeric strings are parsed by `std::strtoull`; the code does not currently validate that the whole token was numeric. The stride is applied by `RunConfiguration::SetStride`. It controls frame dispatch: frame 0 is processed, then every `N`-th TRR frame after that is processed. MOPAC, per-frame NPY emission, and per-frame PDB emission all run on exactly the dispatched frames. There is no current `-npy-stride`, `-pdb-stride`, or `-mopac-stride`.

8.2 Trajectory loader rules

The TPR loader classifies leading non-water, non-ion molblocks as the protein region. Water is a 3-atom moltype with one oxygen and two hydrogens in one residue. An ion is a one-atom, one-residue moltype. Any non-water, non-ion moltype after the protein/solvent region is rejected:

```
Unsupported moltype after protein/solvent region: 'NAME' (atoms=A residues=R nmol=M). Cofactors
    and co-solutes are not currently supported by the trajectory loader.
```

Each protein-shaped molblock must have `nmol=1`. Replicated protein moltypes are rejected:

```
Protein-shape molblock 'NAME' has nmol=M (expected 1; pdb2gmx convention). Replicated protein
    moltypes are not supported.
```

At least one protein-shaped molblock must exist:

```
no protein-shape molblock found in DIR/production.tpr
```

When a parsed `topol.top` readback block is used, every protein residue must have a matching readback entry. Missing entries, out-of-range residue indices, and mismatches between the TPR residue name and the readback comment are hard errors. Without a readback block, residue names are resolved through the naming registry. Unknown residue names are hard errors.

The trajectory protein is finalized from the first TRR frame. Protein positions are converted from nm to Angstrom. Protein velocities, when present, are converted from nm/ps to Angstrom/ps. Solvent water and ion positions are split into the solvent side channel for solvent-aware calculators.

8.3 Trajectory output

Trajectory mode requires `-output DIR`. It writes:

- `DIR/trajectory.h5`, containing trajectory result groups, trajectory frame indices/times, selections, and trajectory protein atoms.
- One top-level topology sidecar set in `DIR`.
- Per-frame PDB files in `DIR/pdbs`. File names are `STEM_fNNNNNN_tT.T.pdb`, where `STEM` is the trajectory directory basename, `NNNNNN` is the raw TRR frame index padded to six digits, and `T.T` is the frame time in ps with one decimal place.
- Per-frame NPY directories in `DIR/npys/frame_NNNNNN`. Each directory receives the same per-conformation feature-array surface that a single-conformation run writes for that frame. `DIR/npys` also receives the per-protein sidecars on the first emitted frame.

Without `-mopac`, the run shape is `PerFrameExtractionSet`: APBS and AIMNet2 run on every dispatched frame, MOPAC is skipped, and vacuum Coulomb is skipped. With `-mopac`, the run shape is `FullFatFrameExtraction`: it inherits the per-frame stack, enables MOPAC on every dispatched frame, enables vacuum Coulomb for the MOPAC-vs-FF14SB reconciliation probe, and adds the MOPAC-family trajectory results.

9 MOPAC behavior by mode

Modes 1–4:

- Default: MOPAC on.
- Meaningful switch: `-no-mopac`.
- APBS stays on.
- AIMNet2 stays on and requires a model.
- Home-rolled vacuum Coulomb stays skipped in these single-conformation modes.

Mode 5:

- Default: MOPAC off.
- Meaningful switch: `-mopac`.
- APBS and AIMNet2 stay on.
- `-mopac` changes the run shape from `PerFrameExtractionSet` to `FullFatFrameExtraction`. That also enables vacuum Coulomb for the FullFat reconciliation result.

10 Out of scope and removed interfaces

These are not supported modes or features in the current five-mode interface:

- propka.
- kaml.
- `-analysis` and `-analysis-h5`.
- `-fleet`.
- `-no-apbs`.
- `-no-coulomb`.
- `-npy-stride`, `-pdb-stride`, and `-mopac-stride`.

The parser's lack of a general unknown-flag rejection pass does not make these interfaces available. If one of these flags appears beside a valid mode, it may be ignored rather than rejected. Prepare commands using only the flags documented above.

11 Discrepancies found during code audit

The guide above follows the current code behavior. The audit found these differences between the canonical five-mode prose, CLI help/comments, and current execution:

- The canonical mode block says MOPAC is an orthogonal `-mopac`/`-no-mopac` toggle across all five modes. Current parsing is not symmetric: modes 1–4 only consume `-no-mopac` and default on; trajectory only consumes `-mopac` and defaults off.
- The canonical mode block says trajectory NPY and PDB emissions have independent strides `m` and `n`. Current code has one `-stride N`; processing, MOPAC when enabled, NPY emission, and PDB emission all use that same dispatched-frame cadence.
- The brief asks for an `.xtc`-rejected validation check. Current validation does not mention `.xtc`. It requires `production.trr`; a directory with only `production.xtc` fails through the missing-`production.trr` file check.
- CLI usage text and `Validate.cpp` mark `NAME_nmr.out` as optional for `-orca` and `-mutant`. Current execution always sets the expanded `_nmr.out` path in `RunOptions`, so a missing file causes ORCA shielding load failure. The operator guidance therefore treats `_nmr.out` as required for a successful mode-3 or mode-4 run as currently wired.
- `-pH` exists in code and help for `-pdb`, with default `7.0`. It is not named in the canonical five-mode block. Current execution records the value but does not pass it into the reduce wrapper.
- Common options `-output`, `-config`, `-aimnet2`, and help flags are real CLI flags but are not part of the mode-selector list in the canonical five-mode block.